

D&D 3.5e Magic Item Enchantment System — Test Documentation

Version: Phases 1-8 Complete

Date: 2026-05-24

Files Modified/Created:

- `EnchantmentType.cs` — 73 enchantment types across weapon, armor, and shield slots
 - `EnchantmentStats.cs` — Data container with 40+ stat fields per enchantment
 - `EnchantmentProperties.cs` — Centralized data-driven registry (no hardcoding)
 - `EnchantmentFactory.cs` — Creation, validation, and pricing logic
 - `EnchantmentEffects.cs` — Runtime combat effect calculations
 - `ItemEnchantmentData.cs` — Per-item instance data (abilities list, Bane type, Defending transfer, etc.)
 - `MagicItemLootGenerator.cs` — CR-based treasure generation (Phase 7)
 - `ItemData.cs` — Updated `GetQualityColor()` and `GetStatSummary()` with enchantment tooltip/color tiers (Phase 8)
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Phase 1-2: Foundation & Core Weapon Abilities

Enchantment Type Registry

#	Enchant- mentType	Slot	Bonus Equiv	Flat Cost	Test
1	Flaming	Weapon	+1	—	✓ +1d6 fire via Bonus- DamageDice/ Type
2	FlamingBurst	Weapon	+2	—	✓ +1d6 fire + crit dice (1d10/2d10/3 d10)
3	Frost	Weapon	+1	—	✓ +1d6 cold
4	IcyBurst	Weapon	+2	—	✓ +1d6 cold + crit dice
5	Shock	Weapon	+1	—	✓ +1d6 electricity
6	Shocking- Burst	Weapon	+2	—	✓ +1d6 electricity + crit dice
7	Corrosive	Weapon	+1	—	✓ +1d6 acid
8	Thundering	Weapon	+1	—	✓ +1d8 son- ic on crit, Fort DC 14 or deaf
9	Holy	Weapon	+2	—	✓ +2d6 vs evil, Align- mentRe- quired=Good
10	Unholy	Weapon	+2	—	✓ +2d6 vs good, Align- mentRe- quired=Evil
11	Axiomatic	Weapon	+2	—	✓ +2d6 vs chaotic, AlignmentRe- quired=Lawf ul
12	Anarchic	Weapon	+2	—	

#	Enchant- mentType	Slot	Bonus Equiv	Flat Cost	Test
					✓ +2d6 vs lawful, AlignmentRe-quired=Chao tic
13	Bane	Weapon	+1	—	✓ +2 enh + 2d6 vs creature type, BaneCreature Type
14	Keen	Weapon	+1	—	✓ DoubleCr- itRange=true , slash/pierce only valida- tion
15	Vorpal	Weapon	+5	—	✓ Re- quiresKeen prerequisite, decapitate on nat 20
16	Vicious	Weapon	+1	—	✓ +2d6 tar- get, SelfDam- age 1d6
17	Wounding	Weapon	+2	—	✓ ConDam- agePerHit=1
18	Speed	Weapon	+3	—	✓ ExtraAt- tackAtFull- BAB=true (haste-like)
19	Throwing	Weapon	+1	—	✓ Al- lowsThrow=t rue, RangeIn- crement=10, melee only
20	Returning	Weapon	+1	—	✓ Return- sAfter- Throw=true,

#	Enchant- mentType	Slot	Bonus Equiv	Flat Cost	Test
					requires ranged/ thrown
21	Distance	Weapon	+1	—	✓ DoubleR- angelIncre- ment=true, ranged only
22	Seeking	Weapon	+1	—	✓ Neg- atesConceal- ment=true, ranged only
23	Defending	Weapon	+1	—	✓ De- fendingACT- ransfer runtime field
24	SpellStoring	Weapon	+1	—	✓ CanStore- Spell=true, MaxStored- SpellLevel=3
25	Merciful- Weapon	Weapon	+1	—	✓ +1d6 non- lethal, MercifulSup- pressed toggle

Phase 5: Advanced Weapon Abilities

#	Enchantment-Type	Slot	Bonus Equiv	Test
26	BrilliantEnergy	Weapon	+4	✓ Ignores armor/shield/natural AC, cannot harm undead/constructs
27	Dancing	Weapon	+4	✓ DancingEffect=true, DancingDuration=4 rounds
28	Disruption	Weapon	+2	✓ Fort DC 14 or destroy undead, BludgeoningOnly validation
29	KiFocus	Weapon	+1	✓ Monk ki abilities through weapon
30	GhostTouch-Weapon	Weapon	+1	✓ Full damage vs incorporeal

Phase 3-4: Armor & Shield Abilities

Core Armor Abilities

#	Enchant- mentType	Slot	Bonus Equiv	Flat Cost	Test
31	Fortification- Light	Armor	+1	—	✓ 25% neg- ate crit/sneak
32	Fortification- Moderate	Armor	+3	—	✓ 50% neg- ate crit/sneak
33	Fortification- Heavy	Armor	+5	—	✓ 75% neg- ate crit/sneak
34	EnergyResist- anceFire	Armor	+1	—	✓ Resist fire 10
35	EnergyResist- anceCold	Armor	+1	—	✓ Resist cold 10
36	EnergyResist- anceElectri- city	Armor	+1	—	✓ Resist elec 10
37	EnergyResist- anceAcid	Armor	+1	—	✓ Resist acid 10
38	EnergyResist- anceSonic	Armor	+1	—	✓ Resist sonic 10
39	ImprovedEn- ergyResist- anceFire	Armor	+2	—	✓ Resist fire 20
40	ImprovedEn- ergyResist- anceCold	Armor	+2	—	✓ Resist cold 20
41	ImprovedEn- ergyResist- anceElectri- city	Armor	+2	—	✓ Resist elec 20
42	ImprovedEn- ergyResist- anceAcid	Armor	+2	—	✓ Resist acid 20
43	ImprovedEn- ergyResist- anceSonic	Armor	+2	—	✓ Resist sonic 20

#	Enchant- mentType	Slot	Bonus Equiv	Flat Cost	Test
44	Great- erEnergyRes- istanceFire	Armor	+3	—	✓ Resist fire 30
45	Great- erEnergyRes- istanceCold	Armor	+3	—	✓ Resist cold 30
46	Great- erEnergyRes- istanceElec- tricity	Armor	+3	—	✓ Resist elec 30
47	Great- erEnergyRes- istanceAcid	Armor	+3	—	✓ Resist acid 30
48	Great- erEnergyRes- istanceSonic	Armor	+3	—	✓ Resist sonic 30
49	Shadow	Armor	+1	—	✓ +5 Hide
50	Improved- Shadow	Armor	+2	—	✓ +10 Hide
51	GreaterShad- ow	Armor	+3	—	✓ +15 Hide
52	SilentMoves	Armor	+1	—	✓ +5 Move Silently
53	ImprovedSi- lentMoves	Armor	+2	—	✓ +10 Move Silently
54	GreaterSi- lentMoves	Armor	+3	—	✓ +15 Move Silently
55	SlickArmor	Armor	+1	—	✓ +5 Escape Artist
56	Improved- Slick	Armor	+2	—	✓ +10 Es- cape Artist
57	GreaterSlick	Armor	+3	—	✓ +15 Es- cape Artist

#	Enchant- mentType	Slot	Bonus Equiv	Flat Cost	Test
58	GhostTouch	Armor	+3	—	✓ Full AC vs incorporeal
59	Invulnerability	Armor	+3	—	✓ DR 5/magic
60	WildArmor	Armor	+3	—	✓ Melds with wild shape
61	SpellResistance13	Armor	+2	—	✓ SR 13
62	SpellResistance15	Armor	+3	—	✓ SR 15
63	SpellResistance17	Armor	+4	—	✓ SR 17
64	SpellResistance19	Armor	+5	—	✓ SR 19

Phase 6: Advanced Armor/Shield Abilities

#	Enchant- mentType	Slot	Bonus Equiv	Flat Cost	Test
65	Glamered	Armor	0	2,700 gp	✓ Disguise Self at will, flat pricing
66	Etherealness	Armor	+4	—	✓ Ethereal Jaunt 1/day, 10 min
67	UndeadControlling	Armor	+3	—	✓ Command undead as evil cleric

Core Shield Abilities

#	Enchant- mentType	Slot	Bonus Equiv	Flat Cost	Test
68	ArrowDeflec- tion	Shield	+2	—	✓ Deflect 1 ranged/round
69	Bashing	Shield	+1	—	✓ +2 size damage in- crease
70	Blinding	Shield	+1	—	✓ Flash 2/ day, Fort DC 14 or blind
71	Animated	Shield	+2	—	✓ Floats, de- fends without holding
72	Reflecting	Shield	+5	—	✓ Reflect spell 1/day
73	GhostTouch- Shield	Shield	+3	—	✓ Block in- corporeal touch attacks

Pricing Verification

Enhancement Bonus Pricing (DMG Table 7-14/7-15)

Enhancement	Weapon Cost	Armor/Shield Cost	Verified
+1	2,000 gp	1,000 gp	✓
+2	8,000 gp	4,000 gp	✓
+3	18,000 gp	9,000 gp	✓
+4	32,000 gp	16,000 gp	✓
+5	50,000 gp	25,000 gp	✓

Effective Bonus Pricing Examples

Item	Enhancement	Abilities	Effective	Weapon Cost	Verified
+1 Flaming Longsword	+1	+1 (Flaming)	+2	8,000 gp + base	✓
+1 Holy Flaming Longsword	+1	+3 (Holy+Flaming)	+4	32,000 gp + base	✓
+1 Vorpal Keen Longsword	+1	+6 (Vorpal+Keen)	+7	98,000 gp + base	✓
+5 Speed Longsword	+5	+3 (Speed)	+8	128,000 gp + base	✓

Maximum Effective Bonus Cap: +10

- Validated in `EnchantmentFactory.CreateEnchantedVariant()` — rejects if total > 10
- Validated in `MagicItemLootGenerator.AddRandomAbilities()` — skips abilities exceeding cap

Flat-Cost Ability Pricing

Ability	Flat Cost	Verified
Glamered	2,700 gp	✓

Combat Effect Tests

Weapon Damage Effects

Effect	Mechanic	Implementation	Verified
Elemental damage (Flaming/Frost/Shock/Corrosive)	+1d6 typed damage	Enchantment - Stats.BonusDamageDice + BonusDamageType	✓
Burst damage (FlamingBurst/IcyBurst/ShockingBurst)	Extra dice on crit	CritBonusDice scaled by multiplier	✓
Alignment damage (Holy/Unholy/Axiomatic/Anarchic)	+2d6 vs alignment	AlignmentDamageDice + AlignmentRequired + AlignmentHelper	✓
Bane	+2 enh +2d6 vs type	BaneEnhancementBonus + BaneDamageDice + BaneCreatureType	✓
Vicious	+2d6 target, 1d6 self	SelfDamage field	✓
Wounding	1 CON/hit	ConDamagePerHit field	✓
Thundering	+1d8 sonic on crit	Deafen Fort DC 14	✓
Disruption	Destroy undead Fort DC 14	CheckDisruptionEffect() in AdvancedEnchantmentEffects	✓

Critical Hit Effects

Effect	Mechanic	Verified
Keen	Double threat range	DoubleCritRange=true , slash/pierce only
Vorpal	Decapitate on nat 20	Requires Keen, immune: no head/immune-to-crit
Fortification (Light/Mod/Heavy)	25%/50%/75% negate crit	FortificationPercentage field

AC & Defense Effects

Effect	Mechanic	Verified
Brilliant Energy	Ignore armor/shield/natural AC	<code>GetBrilliantEnergyACReduction()</code>
Defending	Transfer enh bonus to AC	<code>DefendingACTransfer</code> runtime field
Ghost Touch (Armor)	Full AC vs incorporeal	<code>GhostTouchBonus</code> field
Ghost Touch (Shield)	Block incorporeal touch	<code>GhostTouchShieldEffect</code>
Invulnerability	DR 5/magic	<code>ApplyInvulnerabilityDR()</code>
Animated shield	Defend without holding	<code>IsAnimatedShield()</code>

Energy Resistance

Tier	Amount	Bonus Equiv	Elements	Verified
Base	10	+1	Fire/Cold/Elec/Acid/Sonic	✓
Improved	20	+2	Fire/Cold/Elec/Acid/Sonic	✓
Greater	30	+3	Fire/Cold/Elec/Acid/Sonic	✓

Spell Resistance (Armor)

Enchantment	SR Value	Bonus Equiv	Verified
SpellResistance13	13	+2	✓
SpellResistance15	15	+3	✓
SpellResistance17	17	+4	✓
SpellResistance19	19	+5	✓

SR Check: `CheckCasterLevelVsSR()` — rolls 1d20 + caster level vs SR, combines innate + armor SR via `GetTotalSpellResistance()`.

Skill Bonuses (Armor)

Family	Tiers	Bonuses	Verified
Shadow	Base/Improved/Greater	+5/+10/+15 Hide	✓
Silent Moves	Base/Improved/Greater	+5/+10/+15 Move Silently	✓
Slick	Base/Improved/Greater	+5/+10/+15 Escape Artist	✓

Special Weapon Abilities

Effect	Mechanic	Verified
Speed	Extra attack at full BAB (haste)	ExtraAttackAtFullBAB
Throwing	Melee weapon throwable, 10 ft increment	AllowsThrow , melee-only validation
Returning	Thrown weapon returns	ReturnsAfterThrow , ranged/ thrown validation
Distance	Double range increment	DoubleRangeIncrement , ranged-only validation
Seeking	Negate concealment	NegatesConcealment , ranged-only validation
Spell Storing	Store spell $\leq$ 3rd level, release on hit	CanStoreSpell , MaxStored-SpellLevel=3
Merciful	+1d6 nonlethal, suppressible	MercifulSuppressed toggle
Ki Focus	Monk ki through weapon	KiFocusEffect
Dancing	Fights alone 4 rounds	DancingEffect , DancingDuration=4

Special Shield Abilities

Effect	Mechanic	Verified
Arrow Deflection	Deflect 1 ranged/round	DeflectArrows
Bashing	+2 sizes bash damage	BashDamageIncrease
Blinding	Flash 2/day, Fort DC 14	BlindingFlash
Animated	Float, defend hands-free	AnimatedEffect
Reflecting	Reflect spell 1/day	ReflectingEffect

Special Armor Abilities

Effect	Mechanic	Verified
Wild	Melds with wild shape	WildShapeCompatible
Glamered	Disguise Self at will	GlameredEffect , flat 2,700 gp
Etherealness	Ethereal Jaunt 1/day	EtherealJauntEffect
Undead Controlling	Command undead	UndeadControllingEffect

Validation Rules

Slot Validation

Rule	Verified
Weapon enchantments only apply to weapons	✓ ValidateAbility checks Slot vs ItemType
Armor enchantments only apply to armor	✓
Shield enchantments only apply to shields	✓
Both-slot enchantments (Fortification, Energy Resist) apply to armor AND shield	✓

Prerequisite Validation

Rule	Verified
Keen requires slash or pierce damage type	✓ WeaponDamageTypeRequired
Vorpals requires Keen already applied	✓ RequiredAbility check
Disruption requires bludgeoning weapon	✓ BludgeoningOnly validation
Throwing requires melee weapon	✓ RequiresMeleeWeapon
Returning requires ranged or thrown weapon	✓ RequiresRangedWeapon
Distance requires ranged weapon	✓ RequiresRangedWeapon
Seeking requires ranged weapon	✓ RequiresRangedWeapon

Incompatibility Checks

Rule	Verified
Flaming + IcyBurst OK (different families)	✓
Flaming + FlamingBurst incompatible (same family)	✓ Handled in CheckIncompatibilities
Holy + Unholy incompatible (opposing alignments)	✓
Axiomatic + Anarchic incompatible (opposing alignments)	✓
Total effective bonus capped at +10	✓

Phase 7: Loot Generation

MagicItemLootGenerator Tests

Test Case	Description	Verified
CR budget lookup	<code>GetTreasureBudget(cr)</code> returns DMG Table 7-1 values for CR 1-20	✓
Budget allocation	40-70% of budget to items, rest to coins (via <code>DiceService</code>)	✓
Item type distribution	60% weapon, 25% armor, 15% shield	✓
Enhancement range	1-5 enhancement, limited by budget affordability	✓
Ability selection	Random abilities from <code>EnchantmentProperties.GetForSlot()</code>	✓
Budget constraint	Abilities skip if cost exceeds remaining budget	✓
+10 cap	Abilities skip if total effective bonus would exceed +10	✓
Prerequisite validation	Uses <code>EnchantmentFactory.ValidateAbility()</code> — skips invalid combos	✓
Item cloning	Uses <code>ItemDatabase.CloneItem()</code> — does not modify database originals	✓
Unique IDs	Generated items get unique <code>loot_*</code> IDs	✓
Low budget fallback	Budget < 300 gp returns only gold (no items)	✓
<code>GenerateRandomMagicWeapon()</code>	Single weapon generation API	✓
<code>GenerateRandomMagicArmor()</code>	Single armor generation API	✓
<code>GenerateRandomMagicShield()</code>	Single shield generation API	✓
<code>EstimateItemValue()</code>	Correct pricing for budget tracking	✓

Treasure Value Table (CR → Gold Budget)

CR	Budget (gp)	CR	Budget (gp)
1	300	11	7,500
2	600	12	9,800
3	900	13	13,000
4	1,200	14	17,000
5	1,600	15	22,000
6	2,000	16	28,000
7	2,600	17	36,000
8	3,400	18	47,000
9	4,500	19	61,000
10	5,800	20	80,000

Phase 8: UI Integration

Quality Color Coding (GetQualityColor)

Tier	Effective Bonus	Color	RGB	Verified
Standard	0 (no enchantment)	White	(1,1,1)	✓
Masterwork	0 (MW only)	Light Blue	(0.6, 0.85, 1)	✓
Special Material	0 (material variant)	Purple	(0.7, 0.5, 1)	✓
Uncommon	+1 to +2	Green	(0.2, 0.8, 0.2)	✓
Rare	+3 to +4	Blue	(0.3, 0.5, 1)	✓
Epic	+5 to +7	Purple	(0.7, 0.5, 1)	✓
Legendary	+8+	Orange	(1, 0.5, 0)	✓

Enchantment Tooltip (GetStatSummary → GetEnchantmentTooltipSection)

Feature	Description	Verified
Section header	“— Enchantments —” separator	✓
Ability listing	Each ability with ✧ prefix and display name	✓
Bonus equivalent	[+N equiv] for bonus-priced abilities	✓
Flat cost	[X gp] for flat-priced abilities	✓
Bane creature type	Shows “Bane (Undead)” etc. for Bane weapons	✓
Description truncation	Long descriptions clipped to 80 chars	✓
Total effective bonus	Shows combined enhancement + ability total	✓
Enchanted value	Shows calculated gp price	✓
Auto-integration	Changes flow through InventoryUI.ShowTooltip() automatically	✓
Weapon/Armor/Shield	Tooltip section appears in all three item type blocks	✓

Tooltip Flow

```
InventoryUI.ShowTooltip()
  → item.FullNameWithEnhancement (e.g., "+1 Holy Flaming Longsword")
  → item.GetQualityColor()        (tiered color based on effective bonus)
  → item.GetStatSummary()         (weapon/armor/shield stats)
    → GetEnchantmentTooltipSection() (✧ ability list + descriptions + pricing)
  → item.Description
```

Architecture Compliance

Requirement	Status
Centralized properties (EnchantmentProperties registry)	✓ All data in one place
No hardcoded values	✓ All stats lookup via <code>EnchantmentProperties.Get()</code>
Data-driven combat effects	✓ Effects read stats at runtime
DiceService for all rolls	✓ No <code>System.Random</code> usage
DamageType enum in global namespace	✓ Not using <code>DND35e.Identifiers</code>
ItemData.Id field (not ItemId)	✓
MaterialProperties/ItemMaterial pattern	✓ Follows existing architecture
<code>GetInventoryData().ArmorRobeSlot / LeftHandSlot</code>	✓ Matches inventory slot structure
Fortification: 25%/50%/75%	✓ Per spec
Energy Resistance: +1/+2/+3 bonus equiv	✓ Per spec
Vorpai requires Keen	✓ Per spec

File Summary

File	Lines	Role
EnchantmentType.cs	~90	Enum with 73 enchantment types
EnchantmentStats.cs	~360	Data container: 40+ fields per enchantment
EnchantmentProperties.cs	~600	Centralized registry + lookup API
EnchantmentFactory.cs	~400	Create, validate, price enchanted items
EnchantmentEffects.cs	~300	Runtime combat effect calculations
ItemEnchantmentData.cs	~100	Per-item instance data (abilities, Bane, Defending, Merciful)
MagicItemLootGenerator.cs	~320	CR-based treasure generation
ItemData.cs	+50 lines	GetQualityColor tiers + GetEnchantmentTooltipSection

Total enchantments implemented: 73

Total test cases documented: 100+

All phases (1-8) complete.